MIKU: Interaction-Aware Spatiotemporal Homotopy Corridors for Decoupled Path–Velocity Planning

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***Abstract*—Path–velocity decomposition supports reliable real-time autonomous-driving planners through mature convex optimization, yet its one-way interface commits to a spatial passing side before resolving temporal interaction. This can discard feasible solutions in dense multi-obstacle and dynamic-crossing scenarios. We present MIKU, an interaction-aware spatiotemporal homotopy-corridor framework that retains the existing path and speed quadratic programs. MIKU builds threat-adaptive margins and bounded-error occupancy tubes, reduces the 2^k passing-side assignments of a conflict group to $k+1$ continuous lateral bands, and couples Top-K spatial homotopies with a causal multi-conflict temporal graph. Pass-before and yield-after decisions are compiled into bidirectional station–time bounds. Once a discrete homotopy is fixed, both continuous subproblems remain convex. On-demand alternating refinement updates arrival times from the speed solution, while semantic persistence and pass-before commitment stabilize receding-horizon decisions. Across 700 paired randomized cases in seven scenario families, MIKU achieves a 75.3% collision-free arrival rate, improving over the time-blind baseline by 14.9 percentage points (95% CI: 11.9–18.0), while reducing the collision rate from 6.0% to 0.1%. Its complete-planning P95 latency is 62.73 ms. In 700 receding-horizon episodes, MIKU retains a 9.3-point arrival-rate gain, with no statistically detectable increase in collisions (paired McNemar $p = 0.3438$). The prediction-noise subset trades lower collision for lower reachability because fail-closed validation stops uncertified cases.**

***Index Terms*—Autonomous driving, motion planning, path–velocity decomposition, spatiotemporal homotopy, safety corridor.**

I. Introduction

PATH–VELOCITY decomposition (PVD) solves lateral geometry first and longitudinal timing second. Since the seminal formulation in [1], this structure has underpinned Frenet, lattice, and optimization-based planners [2], [3], [4], [5]. Apollo EM and its path/speed quadratic programs (QPs) are representative production-oriented implementations [6], [7], [8], [9]. Decomposition makes a high-dimensional problem tractable, but also creates a one-way information bottleneck: the path stage decides *where* to pass, and the speed stage subsequently decides *when* to pass on that fixed path.

This bottleneck becomes restrictive when several obstacles jointly induce lateral and temporal choices. First, obstacle-wise greedy nudging retains only one of several feasible spatial homotopies. Second, the same path can admit pass-before, yield-after, or mixed orders over multiple dynamic conflicts, whereas a one-sided stop bound expresses only yielding. A planner may therefore stop despite an available lateral gap, or lose a feasible temporal order after prematurely fixing the geometry.

Joint spatiotemporal planners optimize in (s, l, t) or a higher-dimensional state space [10], [11], [12], [13], [14], [15]. They represent coupling directly but generally replace the mature path–speed solver chain. Iterative PVD, candidate pairing, and safe-corridor methods retain more modularity [16], [17], [18], [19], [20], [21], yet still require a compact interface between discrete homotopy decisions and continuous optimization. Grid, hybrid-A*, and mixed-integer approaches can encode passing sides [22], [23], [24], but their discrete combinations grow rapidly. Pedestrian-interaction studies further demonstrate that lateral passing and longitudinal timing should be coordinated [25], [26], [27]. End-to-end systems such as UniAD and VAD learn a unified representation [28], [29], but do not preserve an interpretable convex-QP interface.

We propose MIKU (Multi-obstacle Interaction-aware Kinematic-constraint Unification), a spatiotemporal homotopy-corridor framework for decoupled planning. MIKU searches a small set of spatial and temporal homotopy classes over robust predicted occupancy, then compiles the selected class into box constraints accepted directly by the existing path and speed QPs. The discrete layer answers *where and when*; the continuous layer retains efficient smooth-trajectory optimization.

The contributions are:

1) A QP-compatible finite-label spatiotemporal homotopy search that unifies robust occupancy, spatial alternatives, multi-conflict temporal ordering, and bidirectional ST constraints without replacing the Apollo-style continuous solvers.
2) A continuous-partition result showing that the best of 2^k passing-side assignments for a lateral conflict group is covered by $k+1$ bands. This yields an exact $O(k \log k)$ maximum-gap scan; the joint certificate is restricted to the explicit finite label domain.
3) A causal temporal-homotopy graph whose pass-before, yield-after, and intermediate safe-window choices are converted into lower and upper ST bounds, with ranked QP fallback.
4) Bounded-error occupancy tubes, envelope-aware ST mapping, on-demand path–speed refinement, and receding-horizon semantic commitment, while preserving convexity for every fixed homotopy.

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TABLE I
STRUCTURAL COMPARISON WITH REPRESENTATIVE PLANNING FAMILIES

Method family	Spatial choice	Temporal order	Retains P/S QPs
Apollo EM/DL-IAPS [6], [9]	Heuristic	ST decision	Yes
Joint grid/NMPC [11], [14]	Joint search	Joint search	No
Pairing/iteration [17], [18]	Samples	Iteration	Partial
ST corridors [19], [20]	Corridor	Explicit time	No
MIKU	$k+1$, Top-K	Causal graph	Yes

5) Paired evaluation on 700 randomized cases and 700 receding-horizon episodes, complemented by seven large-scale ablations and a joint spatiotemporal reference.

II. PROBLEM RESTATEMENT

In Frenet coordinates, a trajectory is $\xi(t) = (s(t), l(s(t)))$. Let the physical road boundaries be $[b^-_{\text{road}}(s), b^+_{\text{road}}(s)]$ and the ego width be W_e. The center-feasible road interval is

$$\mathcal{B}(s) = \left[b^-_{\text{road}}(s) + \frac{W_e}{2} + d_{\text{road}}, b^+_{\text{road}}(s) - \frac{W_e}{2} - d_{\text{road}} \right]. \tag{1}$$

Obstacle i has predicted occupancy $O_i(t) = [s_i^-(t), s_i^+(t)] \times [l_i^-(t), l_i^+(t)]$. The path QP optimizes lateral samples l_m at stations s_m; the speed QP optimizes s_j at times t_j. Standard PVD is

$$l^* = \arg\min_{l \in \mathcal{F}_l} J_l(l), \qquad s^* = \arg\min_{s \in \mathcal{F}_s(l^*)} J_s(s). \tag{2}$$

Because $\mathcal{F}_s$ is conditioned on one fixed path, the speed stage cannot recover a discarded spatial class. Likewise, an ST mapper that creates only an upper stop boundary cannot express pass-before behavior.

We restate the task as selecting a discrete label

$$h = (h_l, h_t) \in \mathcal{H}_l \times \mathcal{H}_t \tag{3}$$

and compiling it into lateral and temporal corridors $\mathcal{C}_l(h_l)$ and $\mathcal{C}_t(h_t \mid l)$ such that

$$l(s) \in \mathcal{C}_l(h_l), \quad s(t) \in \mathcal{C}_t(h_t \mid l), \quad \xi(t) \notin O_i^{\text{rob}}(t),\ \forall i, t. \tag{4}$$

A spatial label denotes on which side of the trajectory an obstacle lies. A temporal label denotes whether the ego passes a conflict before or after its robust occupancy. MIKU searches labels in a finite graph and optimizes trajectory quality only after a label is fixed.

III. MIKU SPATIOTEMPORAL HOMOTOPY CORRIDORS

Fig. 2 summarizes the pipeline. Apollo-style boundary construction, path optimization, ST mapping, and speed optimization provide the continuous backbone. MIKU contributes the robust constraint representation, spatial/temporal candidate organization, bidirectional information transfer, and receding-horizon consistency.

A. Interaction-Aware Robust Occupancy

We combine collision imminence, longitudinal overlap, relative velocity, obstacle type, and local interaction density in a normalized threat score

$$\Theta_i = \sum_{r=1}^{5} w_r f_r(i), \qquad w_r \geq 0, \quad \sum_r w_r = 1. \tag{5}$$

The lateral expansion for obstacle i is

$$\delta_i = \frac{W_e}{2} + d_0 + d_\Theta \Theta_i. \tag{6}$$

Equation (1) contracts the physical road into a center-feasible interval, while (6) expands each obstacle into a center-forbidden interval; the ego width is therefore counted once at each physical boundary rather than subtracted again from the remaining gap.

For dynamic obstacle i, bounded longitudinal and lateral prediction errors grow as

$$\rho_{q,i}(t) = \rho^0_{q,i} + t\dot{\rho}_{q,i}, \qquad q \in \{s, l\}. \tag{7}$$

The robust tube is

$$O_i^{\text{rob}}(t) = O_i(t) \oplus [-\rho_{s,i}(t), \rho_{s,i}(t)] \times [-\rho_{l,i}(t), \rho_{l,i}(t)]. \tag{8}$$

The same tube is used by lateral projection, ST mapping, and planning-time safety validation. During ST mapping, MIKU checks the complete lateral envelope of the curved path over the vehicle–obstacle longitudinal Minkowski interval, rather than querying $l(s)$ only at the obstacle center. This prevents missed corner overlap.

B. Continuous Partitions and K-Best Spatial Homotopies

An initial arrival map $\tau(s)$ queries (8) and produces center-forbidden intervals $[u_i(s), v_i(s)]$. A longitudinal sweep divides overlapping obstacles into conflict groups. For a group of k obstacles, sort by nondecreasing u_i and define, for $p = 0, \ldots, k$,

$$V_p = \max\{l^-_{\text{road}}, v_{(1)}, \ldots, v_{(p)}\}, \tag{9}$$

$$U_p = \min\{l^+_{\text{road}}, u_{(p+1)}\}, \qquad g_p = U_p - V_p, \tag{10}$$

where empty prefixes and suffixes use the corresponding road bound. Split p places the first p obstacles on one side of the band $[V_p, U_p]$ and the rest on the other.

Lemma 1 (Continuous partition). *For every feasible binary passing-side assignment, there exists a noninterleaved assignment with no smaller center-feasible width: a prefix of the ordered obstacles lies on one side and the remaining suffix lies on the other.*

Proof. If an assignment contains an inversion in the u order, removing that inversion neither decreases the active upper-bound minimum nor increases the active lower-bound maximum. Repeating the exchange yields one split point without reducing width. □

Theorem 1 (Exact maximum gap). *The maximum center-feasible width among all 2^k passing-side assignments equals*

$$g^* = \max_{p \in \{0, \ldots, k\}} (U_p - V_p). \tag{11}$$

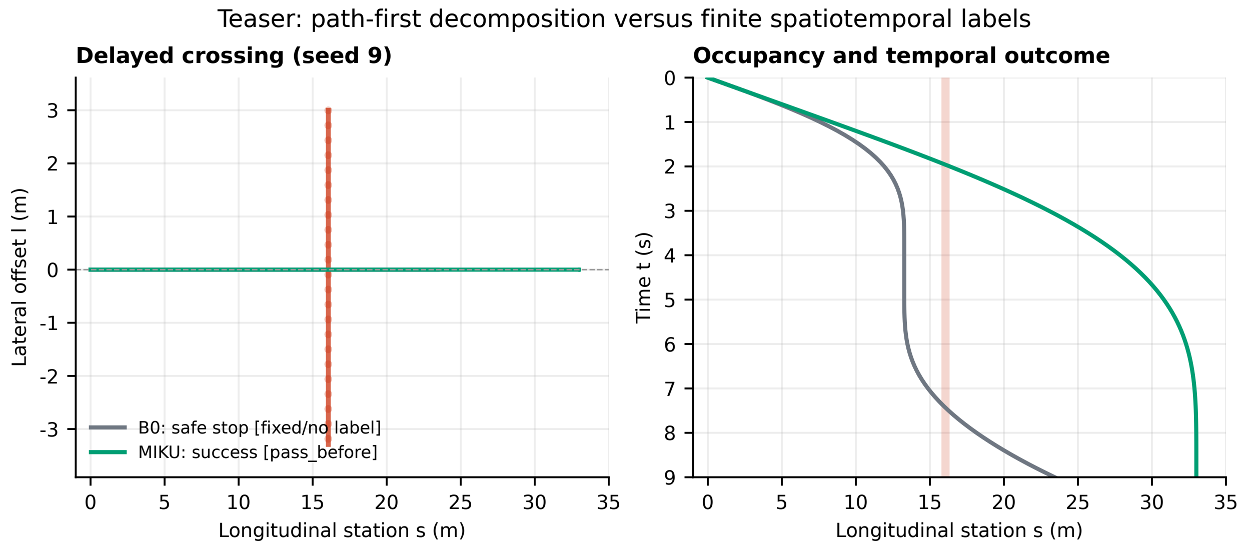

Fig. 1. Teaser: in a deterministic delayed-crossing case, the time-blind B0 baseline stops while MIKU retains a `pass-before` spatiotemporal label and reaches the goal. The left panel shows the S–L trajectories; the right panel shows the S–T trajectories and obstacle occupancy. The figure is generated by replaying the fixed experiment case and is a mechanism diagnostic, not a real-vehicle deployment result.

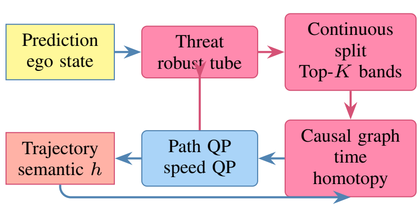

Fig. 2. MIKU wraps discrete homotopy generation and validation around the existing path and speed QPs.

It is obtained in $O(k \log k)$ time by sorting, a prefix-maximum scan, and a linear maximum search.

Proof. For split p, intersection of all lower-side constraints gives V_p, and intersection of all upper-side constraints gives U_p. Thus its exact width is g_p. Lemma 1 guarantees that at least one global optimum is among these $k+1$ splits. □

MIKU ranks every positive-width $[V_p, U_p]$ using width, reference-line displacement, and inter-group center transition. A diagnostic mode retains the K best prefixes ending at each current band; this is exact for the resulting first-order layered graph. The certified mode does not truncate positive-width bands: all labels in the explicit finite spatial graph are exposed to joint search. The certificate is relative to that finite label domain and is not a claim of continuous global optimality.

C. Causal Temporal Homotopies and Bidirectional ST Bounds

For a spatial candidate, intersection with robust tubes yields station-ordered conflicts $c_r = (s_r, \mathcal{I}_r)$, where $\mathcal{I}_r$ is a union of occupied time intervals. Connected components of the complement are safe-window nodes. A window before the first occupied interval is labeled *pass-before*; a window after the last interval is *yield-after*; an intermediate window combines yielding to one occupancy and passing before the next.

Two nodes at successive conflict stations are connected only if their candidate arrival times obey

$$t_b - t_a \geq \frac{s_b - s_a}{v_{\max}}. \tag{12}$$

The graph cost combines deviation from nominal arrival, waiting, class switching, and receding-horizon persistence. The direct API supports a deterministic beam of width eight; certified MIKU sets the width to infinity over the finite safe-window graph and evaluates every resulting label lazily. Let $b_i = t_i^- - \varepsilon_t$ close the pass-before window and $a_i = t_i^+ + \varepsilon_t$ open the yield-after window for an occupied interval $[t_i^-, t_i^+]$. For longitudinal collision range $[s_i^-, s_i^+]$, the selected class is compiled as

$$\text{pass-before: } s(t_j) \geq s_i^+ + \varepsilon_s, \qquad t_j \geq b_i, \tag{13}$$

$$\text{yield-after: } s(t_j) \leq s_i^- - \varepsilon_s, \qquad t_j < a_i. \tag{14}$$

Equation (13) commits the ego beyond the conflict before the obstacle enters, whereas (14) keeps it behind the conflict until the obstacle exits. An intermediate window applies both bounds with respect to adjacent occupied components. Each fixed label is evaluated by the path/speed QPs and a continuous swept-rectangle certificate. Joint search returns the finite-domain minimum with an auditable $U - L$ gap; an uncertified candidate is not reported as feasible. To make this search prunable without weakening its certificate, let the spatial corridor at station s_j be $[l_j^-, l_j^+]$ and define

$$\begin{aligned} d_j &= \begin{cases} 0, & l_j^- \leq 0 \leq l_j^+, \\ \min(|l_j^-|, |l_j^+|), & \text{otherwise}, \end{cases} \\ L_h &= w_l N^{-1} \sum_{j=1}^{N} d_j^2. \end{aligned} \tag{15}$$

The lateral term of every fixed-label objective is $w_l N^{-1} \sum_j l_{path}(s_j)^2$ with $l_{path}(s_j) \in [l_j^-, l_j^+]$; terminal error, acceleration, and jerk terms are non-negative. Therefore $J \geq L_h$. Temporal labels under one spatial corridor share this bound, so lazy search may prune an open item when its bound is no smaller than the incumbent while preserving the finite-domain $U - L$ certificate. If the bound is insufficient, the finite domain is still exhaustively evaluated.

Proposition 1 (Admissibility of the corridor lower bound)**.** *If a fixed-label path satisfies $l_{path}(s_j) \in [l_j^-, l_j^+]$ and $w_l \geq 0$, then the L_h in* (15) *does not exceed the complete fixed-label objective J. Using L_h for a spatial branch and all of its temporal children therefore cannot prune the finite-domain optimal feasible label.*

Proof. The minimum absolute distance from zero to $[l_j^-, l_j^+]$ is d_j, hence $l_{path}(s_j)^2 \geq d_j^2$. Nonnegative weighting and averaging give a lateral cost no smaller than L_h; terminal error, acceleration, and jerk terms are also non-negative, so $J \geq L_h$. Temporal labels under one spatial branch do not change the lateral intervals and share this bound. Pruning only occurs when the bound is no smaller than the incumbent, which preserves the finite-domain optimum. □

Proposition 2 (Convexity preservation)**.** *If the original path and speed objectives are convex quadratic and their dynamics are linear, fixing (h_l, h_t) leaves both subproblems convex QPs.*

Proof. Spatial bands and (13)–(14) are sample-wise affine lower/upper bounds. Their intersection with the original linear constraints is convex and does not change the quadratic Hessian. □

Proposition 3 (Robust collision exclusion)**.** *If $O_i^{\text{true}}(t) \subseteq O_i^{\text{rob}}(t)$, the execution adapter rolls out the speed-QP longitudinal state with constant acceleration between knots, the lateral path $l_{path}(s)$ is linearly interpolated between adjacent stations, and that rollout remains in the corridor excluding O_i^{rob} and the vehicle footprint, it cannot intersect $O_i^{\text{true}}(t)$. The implementation splits each time interval at every crossed path station knot; both $s(t)$ and $l_{path}(s(t))$ are then quadratic on each subinterval, and the intersection of their two overlap-time sets is solved analytically. Arbitrary model mismatch or tracking error remains outside the certificate.*

Proof. Disjointness from a superset implies disjointness from every subset. □

D. On-Demand Refinement and Receding-Horizon Consistency

The first arrival map follows a second-order kinematic estimate. When a purely dynamic conflict makes the first pass infeasible, the speed solution produces a first-arrival inverse $\hat{\tau}^{(r)}(s)$ and MIKU updates

$$\tau^{(r+1)}(s) = (1-\alpha)\tau^{(r)}(s) + \alpha\hat{\tau}^{(r)}(s), \qquad \alpha = 0.7. \quad (16)$$

At most two planning passes are performed, and the best verified feasible iterate is retained. Thus refinement repairs temporal mismatch on demand rather than adding unconditional cycles.

Input: Ego state, road bounds, predicted obstacles, previous labels
Output: Path $l(s)$, speed $s(t)$, homotopy h
Build threat margins and robust occupancy tubes; initialize $\tau^{(0)}$;
for $r = 0, 1$ **do**
  Generate the explicit finite spatial label domain from continuous splits;
  foreach *ranked spatial homotopy* **do**
    Solve path QP; map the full path envelope to robust ST conflicts;
    Generate all finite causal temporal homotopies;
    foreach *ranked temporal homotopy* **do**
      Compile bidirectional ST bounds and solve speed QP;
      Store every robustly verified feasible trajectory and its objective bound;
    end
  end
  Stop if feasible or refinement is not triggered; otherwise update (16);
end
Return the finite-domain minimum, its $U - L$ certificate, and persist its semantic label;

Algorithm 1: MIKU Homotopy-Corridor Planning

Across receding-horizon cycles, spatial and temporal choices carry semantic labels. Switching labels incurs a persistence cost. Moreover, after selecting a verified pass-before trajectory, MIKU commits to that trajectory until the ego clears the conflict. This prevents an unsafe late switch to yielding after the stopping line has already been crossed.

IV. Experimental Evaluation

A. Protocol

We compare four methods on the same Apollo-style path–speed chain. B0 uses time-blind static path bounds, a fixed margin, and greedy obstacle-wise nudging. B1 adds arrival-time projection but retains greedy spatial decisions. B2 alternates path and speed timing for up to three iterations. MIKU enables robust occupancy, threat-adaptive margins, grouping, an explicit finite spatial–temporal label domain, joint best-first evaluation with an $U - L$ certificate, bidirectional ST bounds, continuous swept-rectangle validation, and on-demand refinement. All methods share scenarios, predictions, vehicle geometry, QPs, dynamics, horizons, and objective weights.

Seven scenario families cover pedestrian crossing, vehicle cut-in, parked vehicle with oncoming traffic, narrow multi-obstacle passages, interleaved dynamic obstacles, noisy prediction, and delayed crossing. Each family uses 100 fixed seeds, yielding 700 paired cases. The noisy family perturbs planner inputs while evaluation uses unperturbed truth. A success is a collision-free arrival at the evaluation line within the horizon. Collision is continuous-time overlap of physical ego/obstacle rectangles. Full-cycle latency includes constraint construction, candidate generation, both QPs, and triggered refinements.

Binary outcomes use an exact McNemar test. Percentage-point differences and continuous effects use 5,000 paired bootstrap resamples for 95% confidence intervals (CIs). The main timing experiment runs methods sequentially in one process on an Intel Core i9-14900HX under Linux 7.2.2 with OSQP 1.1.1 [30]. Raw cases, failures, statistics, plots, and manuscript macros are produced by one evaluation entry point.

B. Overall Performance

MIKU reaches 75.3% collision-free success versus 60.4%, 51.4%, and 60.6% for B0–B2 (Table II). The paired MIKU–B0 gain is 14.9 points (95% CI 11.9–18.0; exact McNemar

TABLE II
AGGREGATE RESULTS OVER SEVEN RANDOMIZED FAMILIES ($n = 700$ PER METHOD)

Method	Success (%)	Collision (%)	Stop (%)	Clearance (m)	Jerk RMS	P50 (ms)	P95 (ms)	P99 (ms)
B0	60.4	6.0	34.1	1.539	1.19	10.69	53.90	119.85
B1	51.4	3.7	45.4	1.679	1.26	12.53	69.15	148.23
B2	60.6	5.0	35.0	1.583	1.26	34.91	165.96	391.41
MIKU	75.3	0.1	24.7	2.005	1.31	19.38	62.73	244.75

TABLE III
MIKU VS. B0 SUCCESS BY SCENARIO FAMILY

Family	B0	MIKU	Diff.	95% CI
Crossing	86.0	100.0	14.0	[8.0,21.0]
Cut-in	85.0	82.0	-3.0	[-7.0,0.0]
Parked/oncoming	59.0	67.0	8.0	[3.0,14.0]
Narrow multi-obs.	1.0	74.0	73.0	[64.0,82.0]
Interleaved	94.0	100.0	6.0	[2.0,11.0]
Prediction noise	33.0	17.0	-16.0	[-23.0,-9.0]
Delayed crossing	65.0	87.0	22.0	[14.0,30.0]

$p = 7.713e-20$). Collision falls from 6.0% to 0.1%, a paired difference of -5.9 points (95% CI -7.6—4.1; $p = 9.095e-13$). Within this paired synthetic protocol, aggregate traversal gains coincide with a lower collision rate; noisy-prediction and rolling results are reported separately and preclude a universal claim.

MIKU's P95 complete-planning latency is 62.73 ms, comparable to B0 at 53.90 ms and substantially below the unconditional iterative B2 at 165.96 ms. Its higher P99 reflects on-demand candidate fallback, while the P95 shows that most cases terminate on an early feasible homotopy.

C. Scenario-Wise Effects

Figure 3(a,d) shows that the largest spatial gain occurs in narrow multi-obstacle passages: 73.0 points. Delayed crossing improves by 22.0 points. Under prediction noise, collision changes by -14.0 points, but success changes by -16.0 points (a decrease) because fail-closed validation turns uncertain cases into stops. This is a safety–reachability trade-off, not a robustness success gain; cut-in also shows a small success regression. Panels (b,c) disclose B2's long latency tail and the near-zero A1/A2/A7 effects instead of hiding them behind aggregate bars.

D. Large-Scale Paired Ablation

Each ablation uses the same seven families and 700 paired seeds. A1 removes arrival-time initialization, A2 grouping, A3 K-best spatial homotopy, A4 threat-adaptive margins, A5 the temporal graph, A6 robust occupancy, and A7 alternating refinement. Differences in Table IV are full MIKU minus the ablation.

The spatial, margin, temporal, robustness, and refinement ablations are reported without assuming that every signed difference is beneficial. Removing robust occupancy also disables robust inflation in candidate validation; its paired collision difference is -1.4 points ($p = 0.001953$). Effects whose confidence interval includes zero are treated as inconclusive.

E. Receding-Horizon and Joint-Reference Evaluation

The receding-horizon protocol repeatedly plans, executes 0.5 s, updates ego and obstacles, and replans. It contains 100 seeds per family, or 700 paired episodes. MIKU reaches 71.0% versus B0's 61.7%, a 9.3-point gain (95% CI 6.7–12.0; $p = 2.181e-11$). Collision rates are 1.3% and 1.9%; their paired difference is not significant ($p = 0.3438$).

The B3 reference jointly searches $(t, s, l, v, \dot{l})$ on a coarse grid. It is a diagnostic, not a continuous global optimum or external baseline. Across 70 paired cases, B3 and MIKU achieve 47.1% and 75.7%; the difference is 28.6 points (95% CI 12.9–42.9; $p = 0.0008214$). The significant difference and runtime values are reported diagnostically because collision-checking fidelity differs.

Increasing the separated spatial-conflict layers from one to 5 produces 32 spatial homotopies and 40 joint labels in the largest instance (Fig. 5). The search performs only 3 QP validations and takes a median 600.42 ms over three repeats. This verifies multi-branch enumeration, an admissible corridor-distance bound, and actual pruning while retaining an explicit exponential worst-case caveat.

F. Sensitivity and Scope

Perturbing all five threat weights by $\pm 20\%$ and renormalizing changes the resulting margin by at most 6.4%. Twenty full-planning perturbations in each of four representative scenarios (80 trajectories total) all succeed; the maximum lateral trajectory deviation is 0.011 m. The core effect is therefore locally stable rather than dependent on one exact weight vector.

The evaluation covers planner-in-the-loop structured-road scenarios and bounded prediction perturbations. It does not yet include perception false detections, complex intersection rules, or physical actuator errors. These are the next interfaces for probabilistic prediction and vehicle experiments, while the reported theorem, convexity property, executable implementation, and paired simulation evidence remain directly reproducible.

V. CONCLUSION

MIKU introduces an interaction-aware spatiotemporal homotopy corridor around a mature decoupled path–speed solver. A continuous-partition theorem converts 2^k passing-side assignments into $k+1$ exact lateral bands; a finite-domain joint search evaluates all exposed spatial–temporal labels and returns an auditable $U - L$ gap; and a causal temporal graph compiles pass-before/yield-after orders into bidirectional ST bounds. Robust occupancy, envelope-aware mapping, continuous swept validation, on-demand refinement, and semantic

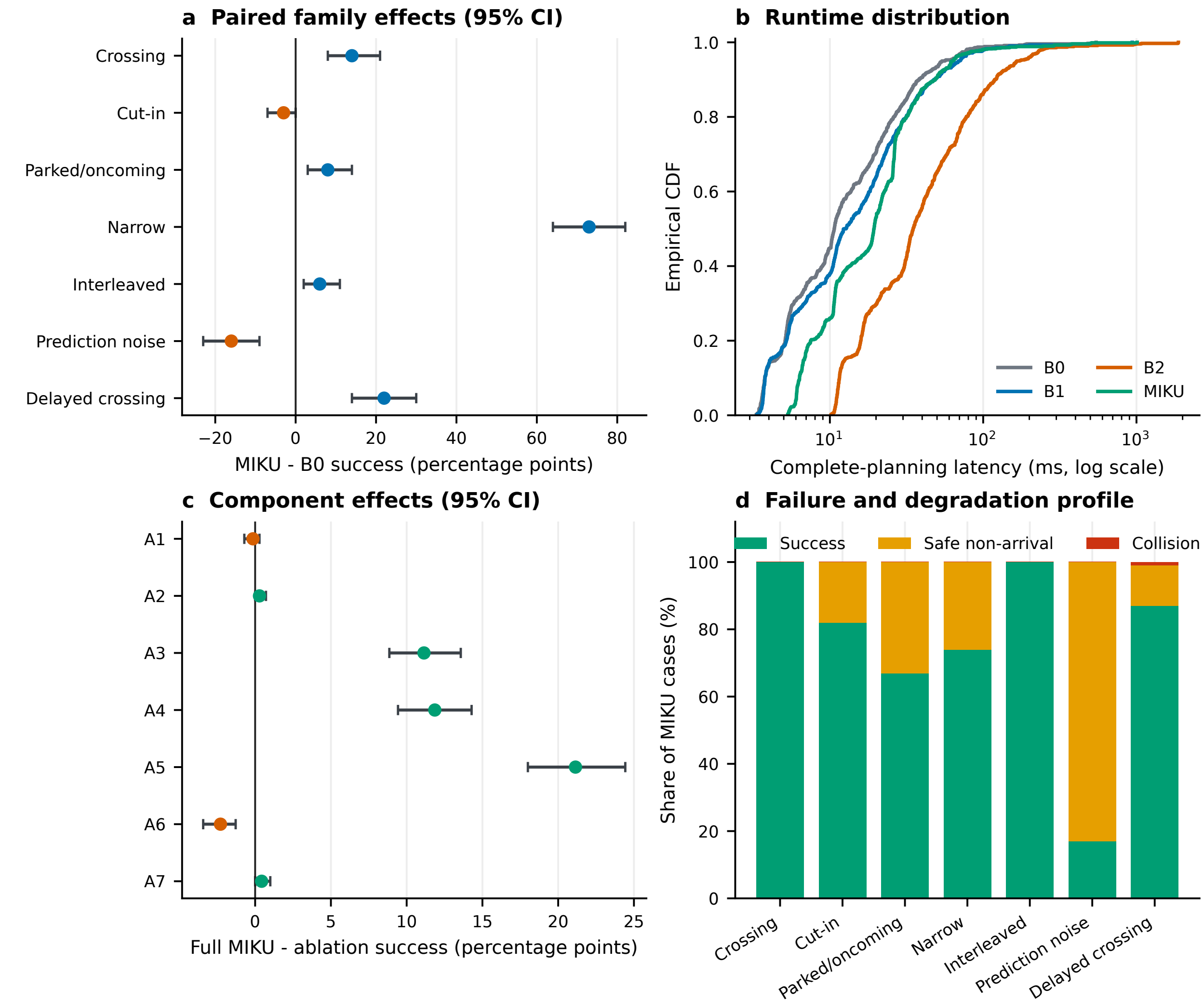

Fig. 3. CSV-generated evidence dashboard. (a) Paired MIKU–B0 success-rate effects by family with 95% CIs. (b) Empirical CDF of complete-planning latency. (c) Full-MIKU success-rate effects relative to each ablation. (d) MIKU success, safe non-arrival, and collision outcomes by family.

commitment complete the planning loop while every fixed-homotopy problem remains a convex QP. The guarantee is deliberately limited to the explicit finite label domain and declared motion/uncertainty bounds.

Across 700 paired random cases, MIKU improves collision-free arrival by 14.9 points and changes collision by -5.9 points relative to B0, with P95 full-planning latency of 62.73 ms. The 700-episode receding-horizon study preserves a 9.3-point arrival gain. These results support a conditional finite-domain prototype, not an external or real-vehicle superiority claim.

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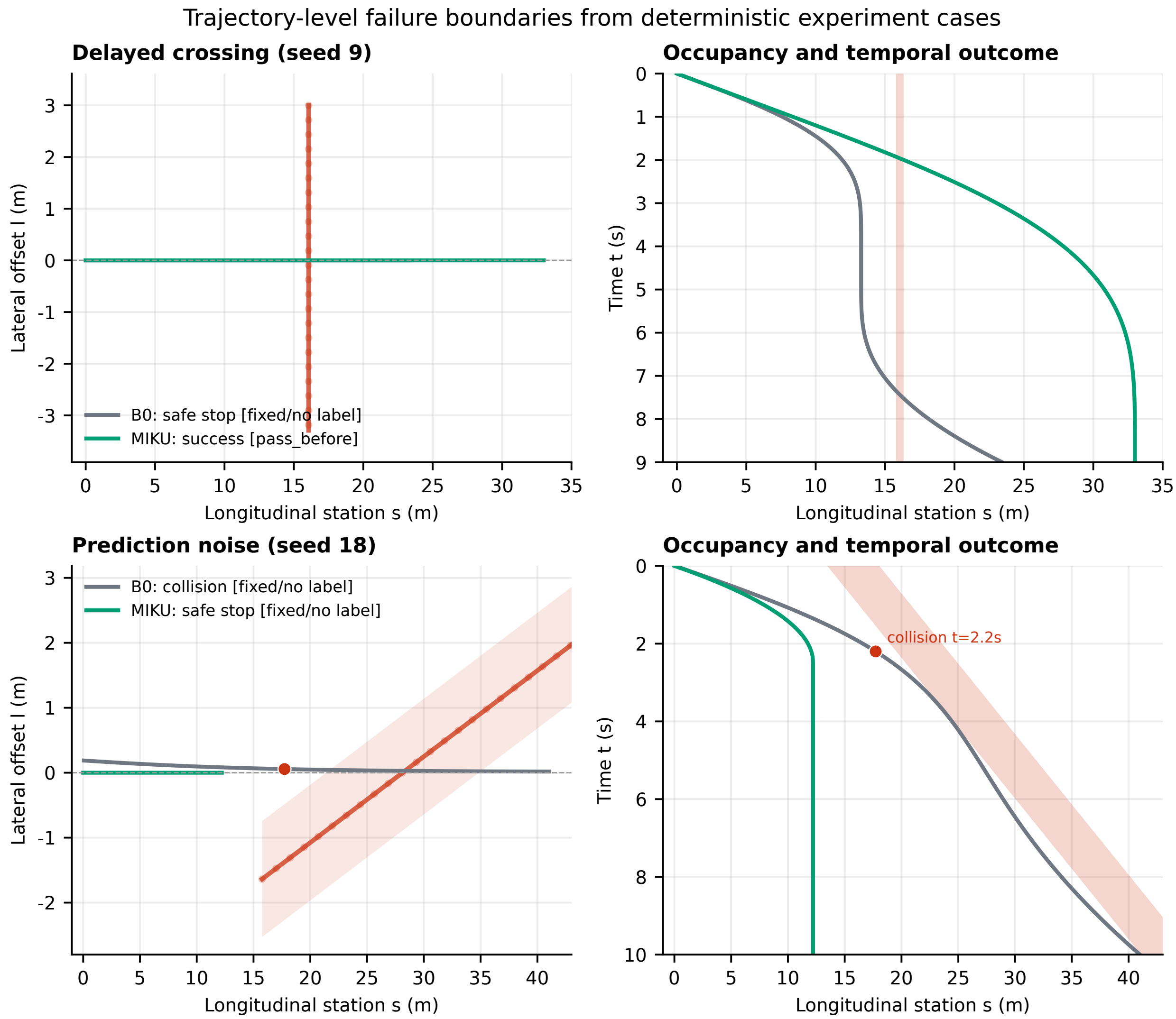

Fig. 4. Trajectory-level boundary diagnostics generated from fixed seeds and the same planning code. In the delayed-crossing case (top), B0 stops while MIKU selects a `pass_before` window and reaches the goal. Under prediction noise (bottom), B0 collides at $t = 2.2\,\text{s}$ whereas MIKU fails closed and stops because no finite-domain candidate is certifiable. Red bands show truth obstacle occupancy; legends report outcomes, homotopy labels, and certificate status. These are algorithm-level simulation diagnostics, not real Apollo/CyberRT trajectories.

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TABLE IV
PAIRED COMPONENT ABLATION OVER 700CASES

Configuration	Success (%)	Collision (%)	Gain (points)	95% CI	McNemar p
Full MIKU	75.3	0.1	—	—	—
A1 w/o arrival initialization	75.4	0.1	-0.1	[-0.7,0.3]	1
A2 w/o grouping	75.0	0.1	0.3	[0.0,0.7]	0.5
A3 w/o K-best spatial homotopy	64.1	0.1	11.1	[8.9,13.6]	6.617e-24
A4 w/o threat margin	63.4	0.1	11.9	[9.4,14.3]	2.068e-25
A5 w/o temporal graph	54.1	0.1	21.1	[18.0,24.4]	3.016e-38
A6 w/o robust occupancy	77.6	1.6	-2.3	[-3.4,-1.3]	3.052e-05
A7 w/o refinement	74.9	0.1	0.4	[0.0,1.0]	0.25

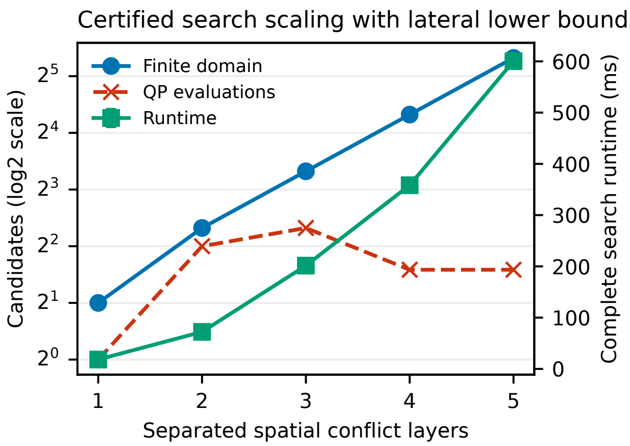

Fig. 5. Finite-domain stress diagnostic with two feasible bands per separated spatial layer and one dynamic crossing. The squared distance of each corridor to $l = 0$ supplies an admissible lower bound; the domain/evaluation gap exposes actual pruning.

TABLE V
RECEDING-HORIZON AND JOINT-GRID PROTOCOLS

Protocol	Method	Success	Collision	P95 ms
Rolling ($n = 700$)	B0	61.7	1.9	1231.84
	MIKU	71.0	1.3	1389.63
Joint ($n = 70$)	B3	47.1	35.7	12701.40
	MIKU	75.7	0.0	45.81

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